

Drona Workflow Engine

Marinus Pennings
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High Performance
Research Computing
DIVISION OF RESEARCH



Historical Background

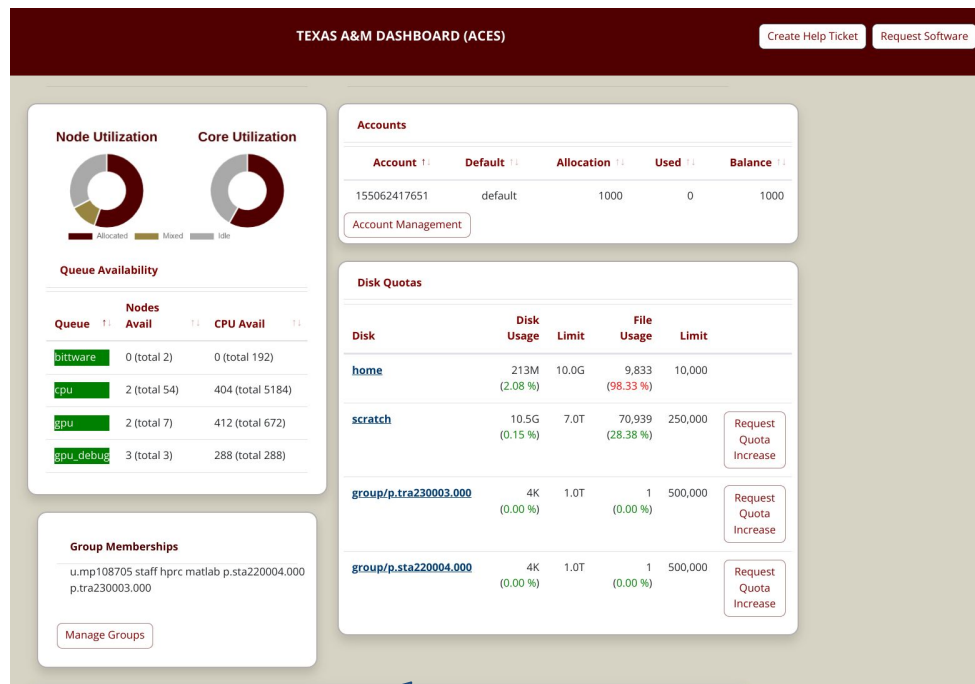
HPRC helpesk receives lots of support tickets

- Many requests; quota increase, groups, software requests
- Users don't always provide all needed information in a help ticket
- Not all users familiar with linux shell, making it hard to use command line tools to check quotas, accounts, job stats, etc
- Lots of tickets asking for help in submitting jobs
 - no experience with linux/shell,vi, etc
 - not familiar with schedulers (e.g Slurm), creating/submitting jobs, loading modules, etc

Be proactive in assisting users to create/submit jobs and manage their resources



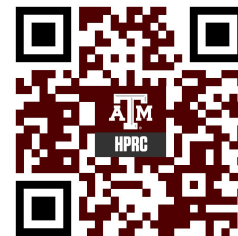
Our Original User Dashboard



TAMU Dashboard:

- graphical, color coded, elements to check quotas, accounts, group membership
- elements to show cluster and queue stats
- guided forms to request help, quota, groups, software, etc

PEARC22 paper introducing dashboard and composer



<https://dl.acm.org/doi/abs/10.1145/3491418.3535182>

Includes some more recent features



Dashboard 2.0 Design Principles

Motivation:

- Every user is unique; most dashboards are not
 - Users should be able to pick and choose dashboard elements
 - Users should be able to move elements around
- Every element should be independent
 - Easier to manage
 - More opportunities to collaborate through sharing
- Focus on user experience



Dashboard 2.0



ACES Dashboard

SettingsRequest Help

Project Information ⓘ

Account ⓘ	FY ⓘ	Default ⓘ	Used / Allocated (%) ⓘ	Balance ⓘ	PI ⓘ
154669186753	2025	☒ Y	41439.87 / 801451.00 (5.17%)	760011.13	Perez, Lisa
155062417651	2025	Set as Default	3221375.74 / 9010994.00 (35.75%)	5789618.26	Liu, Honggao
156171559762	2025	Set as Default	96.51 / 50000.00 (0.19%)	49903.49	Nite, Sandra

Your Jobs ⓘ

No active jobs.

Queue Availability ⓘ

Queue ⓘ	Core Usage (%) ⓘ	Node Usage (%) ⓘ
cpu*		
gpu		
gpu_debug		
pvc		
bitware		

Quota Information ⓘ

Disk ⓘ	Disk Usage (%) ⓘ	File
/home/u.mp108705	602M/10.0G 5.88%	6960/10000 69.60%
/scratch/user/u.mp108705	67.0G/7.0T 0.93%	154675/250000 61.87%
/scratch/group/p.tra230003.000	4K/1.0T 0.00%	1/500000 0.00%

Virtual Env Management ⓘ

Name	Python Version	GCCcore Version	Description	Toolchain	Action
mynextenv	Python/3.12.3	GCC/13.3.0	testing the -p flag		Delete
huggingface	Python/3.10.8	GCCcore/12.2.0	virtual env to interact with HuggingFace		Delete
mypythonenv	Python/3.11.3	GCCcore/12.3.0	This env is just for testing		Delete

User Groups ⓘ

Group ⓘ	Disk Path ⓘ
u.mp108705	/home/u.mp108705
u.mp108705	/scratch/user/u.mp108705

Add Dashboard Elements

Search elements...

[Filter](#) [All](#) [Analysis](#) [System](#) [User](#)

[Node Utilization](#)
Click and drag to dashboard
Displays the current and available computing resources across nodes.

[Python View Manager](#)
Click and drag to dashboard
Manage Python virtual environments directly from the dashboard.

[Quota Information](#)
Click and drag to dashboard
Shows disk quota usage and limits for users and groups.

[User Groups](#)
Click and drag to dashboard
Shows disk quota usage and limits for users and groups.

[Accounts](#)
Click and drag to dashboard

[User Jobs](#)
Click and drag to dashboard

Save/load preferred layouts

Pick what elements to add to dashboard

delete, resize, drag and drop elements

request assistance



Our Original GUI Composer

The screenshot displays the Terraform GUI Composer interface in a web browser. The interface is split into two main sections: 'Job Configurations' on the left and 'Preview' on the right.

Job Configurations:

- Runtime:** A dropdown menu set to 'Matlab'.
- Module:** An empty text input field with an 'Add' button below it.
- Executable Script:** A section with a 'Choose File' button and a text area containing 'matlab-script.m'. Below this, it says 'You must upload an executable script to continue.'
- Commands:** A text area containing 'matlabsubmit matlab-script.m'.
- Job Name:** An empty text input field.
- Walltime:** Three input fields for 'days', 'hours', and 'mins'.
- Request a GPU:** A checkbox that is currently unchecked.
- Total CPU cores:** An input field with the value '1'. Below it, a note says 'If blank, total CPU cores default to 1.'
- CPU cores per node:** An input field with the value '1'. Below it, a note says 'Restriction: maximum of 28 cores per node. If blank, CPU per node equals to total CPU cores.'
- Total:** An input field with the value '120' and a unit dropdown menu set to 'MB'.

Preview:

The preview window shows the generated Terraform configuration. It starts with 'Default values: Time: 2 hours, Total cores: 1, Total cores per node: 1'. Below this is a shell script snippet:

```
#!/bin/bash
##ENVIRONMENT SETTINGS
#SBATCH --export=NONE
#SBATCH --get-user-env=L

#SBATCH --job-name=job_name
#SBATCH --time=2:00
#SBATCH --ntasks=1
#SBATCH --ntasks-per-node=1
#SBATCH --mem-per-cpu=120M

#SBATCH --error=$SCRATCH/job_composer/job_name/job_error.%j
#SBATCH --output=$SCRATCH/job_composer/job_name/job_name.%j

module purge

## Load your requested modules

module load Matlab/R2016b

## Go to the directory where we put the script
cd $SCRATCH/job_composer/job_name

## Ensure job script is unix compatible
dos2unix matlab-script.m

## Run the program
matlabsubmit matlab-script.m
```

Screenshot from
April 16, 2021

Form and preview window, side by side. User enters values in the form and the composer fills in the values right away in the preview window (color coded) Modules automatically added using search bar to avoid typos. Additional features for commonly used Software.

Why Drona Workflow Engine

Our GUI composer did a good job in assisting users to create and submit their jobs, but

- limited number of available form elements, mostly static
- limited user control in creating their own forms and job scripts

At the same time HPC clusters becoming more complex

- Variety of accelerators (GPU, PVC, IPU, Vector Engines)
- Various execution environments (containers, venvs)
- Different schedulers (e.g Kubernetes)

Researchers should only focus on doing their research (but at the same time have the option to create their own personalized workflows)



Drona Workflow Engine

What features did we want?

- Rich set of form elements to guide researcher to provide info for their workflows
- Feedback for user-provided input when needed
- Auto generate as much workflow code as possible
 - Setup computing environments
 - Create boilerplate code
 - Select resources (e.g. type of accelerator)
- USER focus: researchers should be in full control of workflows
 - import relevant workflows
 - Create their own workflows, adapt existing ones
 - Share workflows with their peers
- Ability to integrate backend tools



Drona Example Form

Select run environment

AlphaFold3 specific fields

Dynamic Dropdown to select account

AlphaFold specific info with links

Job Name: DronaAlphaFoldRun

Location: [Change](#) /scratch/user/u.mp108705/drona_composer/runs/DronaAlphaFoldRun

Environments: AlphaFold [+](#)

AlphaFold version: ☒ AlphaFold3 ☐ AlphaFold2

To download the AlphaFold3 model file, you have to agree to the terms of use.
Please fill out the [AlphaFold3 Google Form](#) to request the model file

JSON input [?](#) Directory to model file [?](#) Number of recycles (default: 3) [?](#)

[select](#) [Select](#) 3

Template Date (default today) [?](#) Accelerator [?](#)

-- Choose an option -- [v](#)

Project Account [?](#) Expected time (cpu part) [?](#) Expected time (gpu part) [?](#) Email notification options [?](#)

-- Choose an option -- [v](#) Days Hours Minute Days Hours Minute -- Choose an option -- [v](#)

154669186753 (760011.13)
155062417651 (5789618.26)
156171559762 (49903.49)

[Preview](#) [Show History](#)

Cautions: Job files will overwrite existing files with the same name. The same principle applies for your executable scripts.



Drona Example Preview

Job Preview

Files: driver.sh alphafold3-cpu.job **alphafold3-gpu.job** alphafold3-input.json

Notes:

- Available nodes with gpus: H100=1, A30=1 -> selecting H100
- Drona configures the memory to 64GB. For longer sequences, > 1000 residues, you may need to increase the memory to 128GB. To do so, update the line #SBATCH --mem=64G in the alphafold3-cpu.job tab.

```
1 #!/bin/bash
2
3 #SBATCH --job-name=alphafold3-gpu
4 #SBATCH --time=24:00:00
5 #SBATCH --nodes=1
6 #SBATCH --ntasks-per-node=1      # alphafold3 runs 4 x jackhmmer processes
7 #SBATCH --cpus-per-task=8        # alphafold3 recommends using a max of 8 cpus
8                                 # per jackhmmer process
9 #SBATCH --mem=64G                # minimum of 64GB recommended
10 #SBATCH --partition=gpu --gres=gpu:h100:1
11 #SBATCH --output=stdout.%x.%j
12 #SBATCH --error=stderr.%x.%j
13
14 module purge
15 module load AlphaFold3/3.0.1
16
17 protein_name=$(grep name alphafold3-input.json | cut -f 4 -d '"' | sed 's/_/_/g')
18 lc_protein_name=${protein_name,,}
19
20 AF3_MODEL_PARAMETERS_DIR=
21 AF3_DATABASES_DIR=/scratch/data/bio/alphafold3/db/2025.03.13
22 AF3_CODE_DIR=/scratch/data/bio/alphafold3/code/alphafold3-3.0.1
23 AF3_IMAGE=/sw/hprc/sw/bio/containers/alphafold/alphafold_3.0.1_unlhcc.sif
24
25 AF3_OUTPUT_DIR=$PWD/output_${protein_name}_20250826-235934
26 AF3_INPUT_DIR=$AF3_OUTPUT_DIR/$lc_protein_name
```

Tip: Configure your job on the left, then submit to see live output on the right. Drag the center divider to resize panes.

Submit Job Close

Live Output

READY

Live Output Stream

Job execution output will appear here in real-time once you submit your configuration.

Generated scripts

Immediate feedback

Fully editable preview pane, with full syntax highlighting

Output console



Typical Form Composer

FORM

user enters values for
provided fields in GUI form
and form stores them in
variables. E.g.

nodeV, taskV

nodeV, taskV

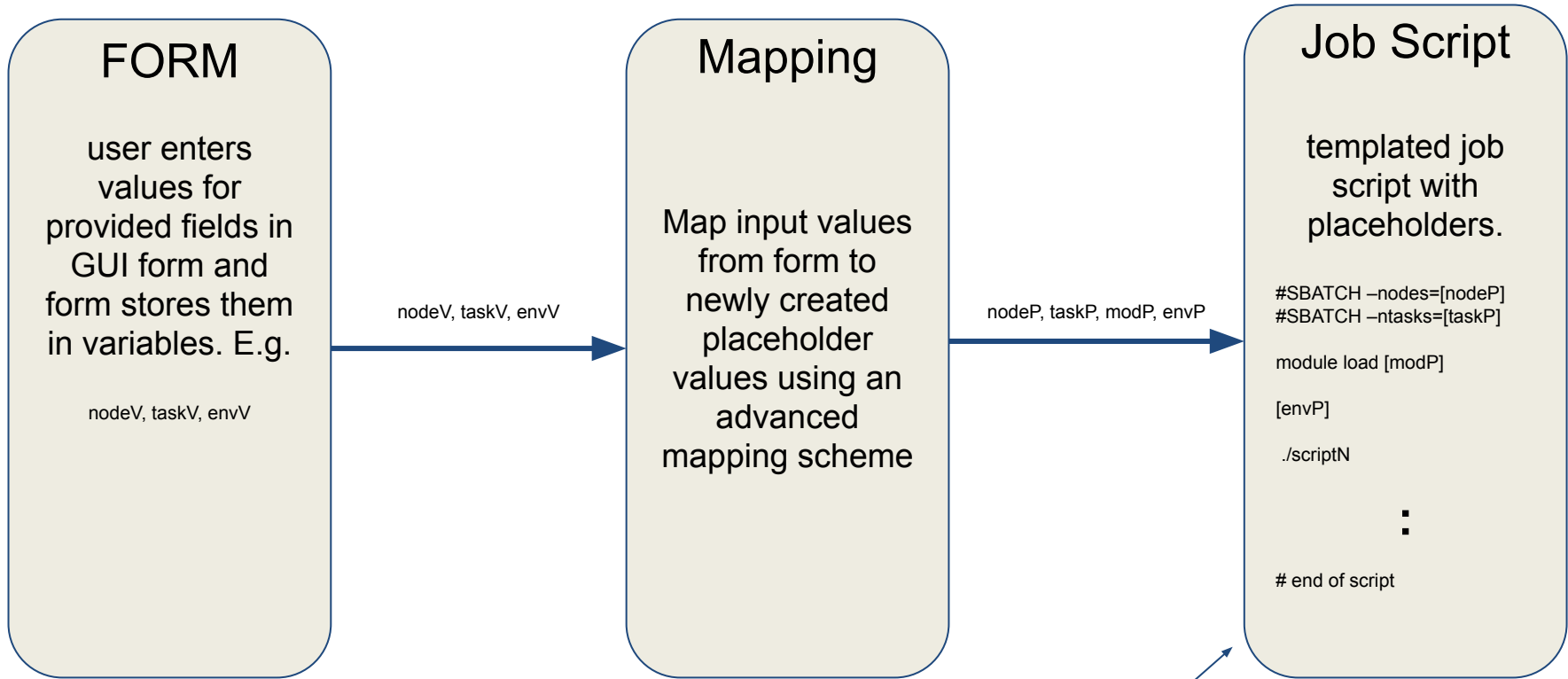
(simple variable substitution)

JOB SCRIPT

templated job script with
placeholders for form values.
E.g.

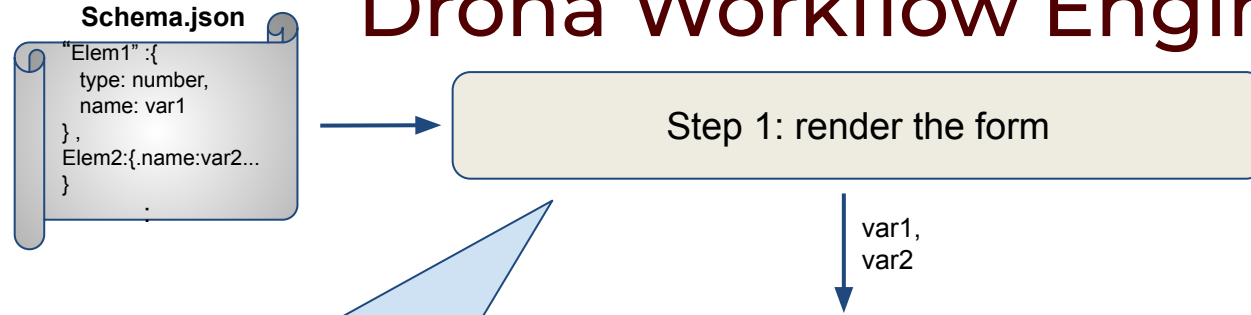
```
#SBATCH --nodes=<nodeV>  
#SBATCH --ntasks=<taskV>
```

Drona Engine



Can be multiple files

Drona Workflow Engine



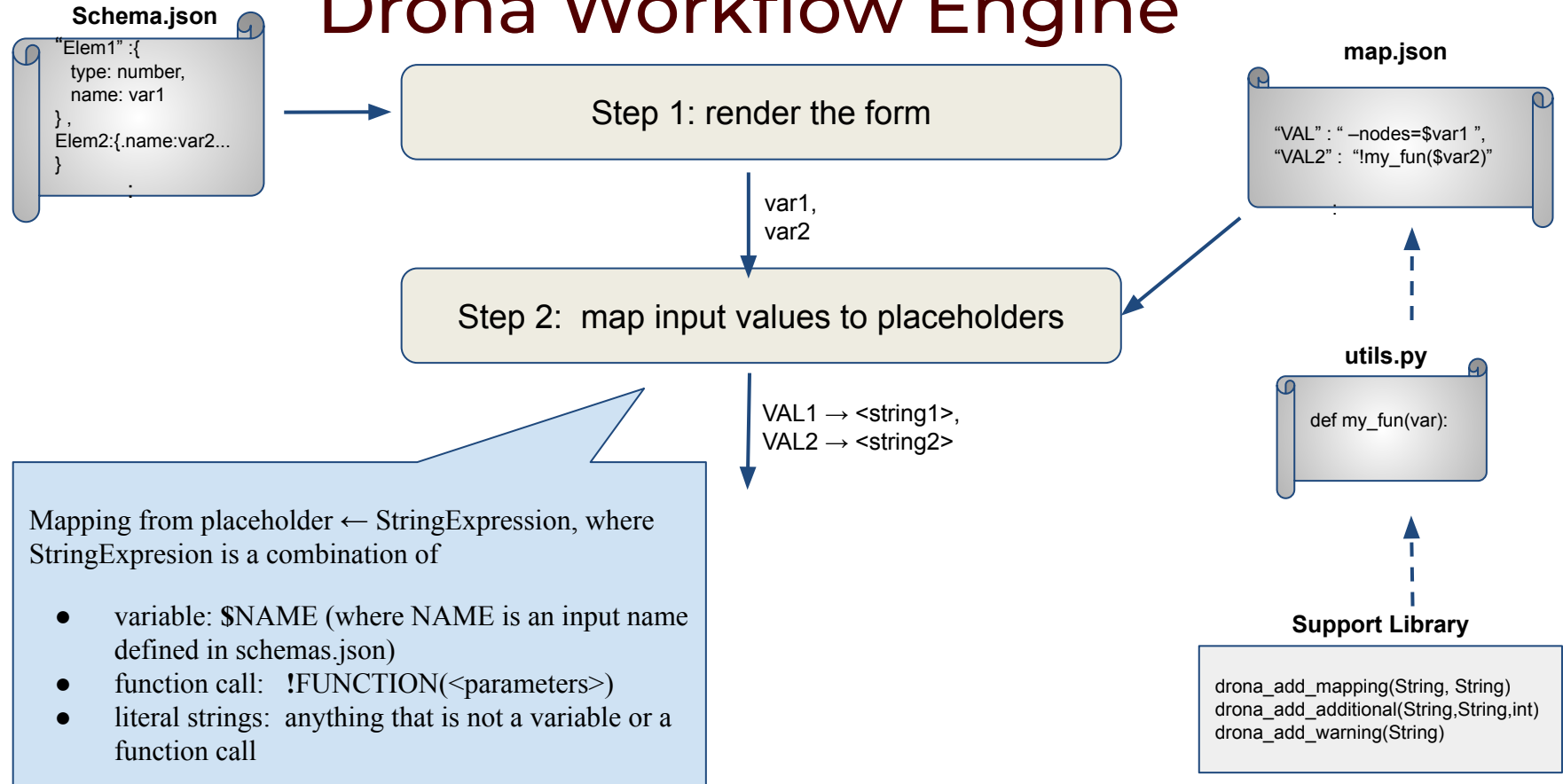
Sample form elements

- number / text
- select / radioGroup / checkbox
- **dynamicSelect**
- etc

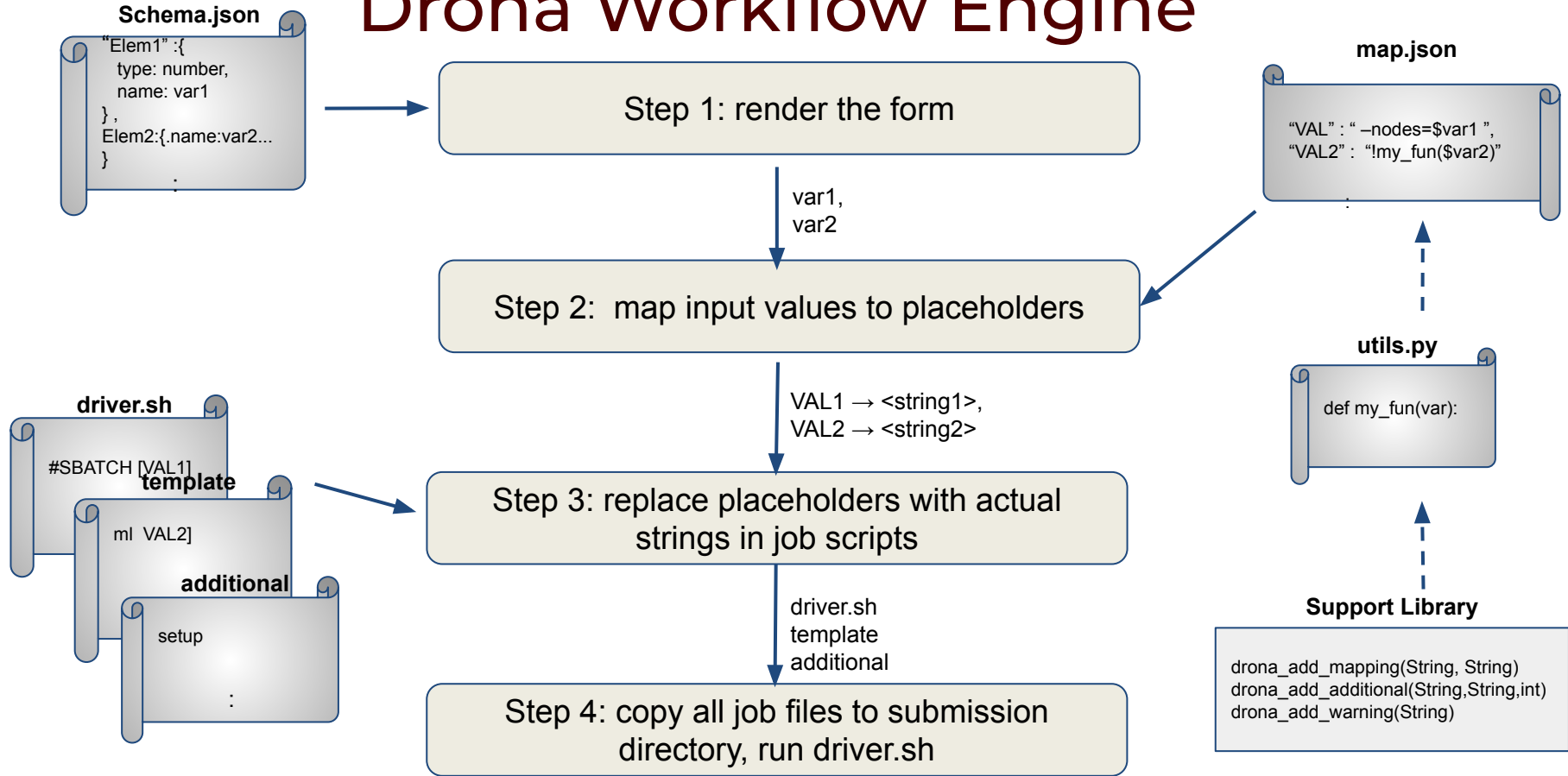
Powerful logic to decide under what conditions form will be shown

- operators !, ||, &&, ()
- use "condition" field in form element

Drona Workflow Engine



Drona Workflow Engine



More information

<https://u.tamu.edu/drona-wfe>



<https://u.tamu.edu/drona-wfe>



<https://github.com/tamu-edu/dor-hprc-drona-composer>

How can you help

- Looking for collaborations
- Contact us for help creating advanced workflows
- Share your workflows
- Any other feedback

contact us at help@hprc.tamu.edu (or through the ACES Dashboard)



ACCESS primer: Intro to Drona workflows
Friday, April 4, 2025 1:30PM-2:30PM CT, Online
https://u.tamu.edu/drona_primer



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Thank you

